Amir Beck, Chapter 1 - Exercise 1.4

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1.4. Prove the Cauchy Schwarz inequality. Show that equality holds if and only if the vectors x and y are lienarly dependent.

Suppose $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent, that is, there exists a scalar $c \in \mathbb{R}$ such that $\mathbf{x} = c\mathbf{y}$, then

$$\begin{aligned} |<\mathbf{x},\mathbf{y}>| &= |<c\mathbf{y},\mathbf{y}>| \\ &= |c<\mathbf{y},\mathbf{y}>| \\ &= |c|.||\mathbf{y}||^{\mathbf{2}} \\ &= (|c|||\mathbf{y}||)||\mathbf{y}|| \\ &= ||c\mathbf{y}||.||\mathbf{y}|| \\ &= ||\mathbf{x}||.||\mathbf{y}|| \end{aligned}$$

Suppose that, without loss of generality, $\mathbf{x} = 0$. Then $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent and by the previous argument the (in)equality is satisfied.

We now suppose that both vectors are nonzero. Define $p : \mathbb{R} \to \mathbb{R}$ such that

$$\begin{aligned} p(t) :&= ||\mathbf{x} + t\mathbf{y}||^2 \\ &=<\mathbf{x} + t\mathbf{y}, \mathbf{x} + t\mathbf{y}> \\ &=<\mathbf{x},\mathbf{x}> +2t<\mathbf{x},\mathbf{y}> +t^2<\mathbf{y},\mathbf{y}> \\ &= ||\mathbf{x}||^2 + 2t<\mathbf{x},\mathbf{y}> +t^2||\mathbf{y}||^2 \end{aligned}$$

Since $\mathbf{y} \neq 0$, $||\mathbf{y}|| \neq 0$ and thus, p is a polynomial of degree 2.

By the positive definiteness property of the inner product, it must be the case that $p(t) \geq 0$, for any $t \in \mathbb{R}$. That implies that

$$\begin{gathered} \Delta = (2<\mathbf{x},\mathbf{y}>)^2 - 4||\mathbf{x}||^2.||\mathbf{y}||^2 \leq 0 \\ \Leftrightarrow \\ <\mathbf{x},\mathbf{y}>^2 \leq ||\mathbf{x}||^2.||\mathbf{y}||^2 \end{gathered}$$

and the result follows by taking square roots on both sides of the previous inequality.

Finally, for the equality to happen, we need

$$0 = \Delta \Leftrightarrow < \mathbf{x}, \mathbf{y} >^2 = ||\mathbf{x}||^2.||\mathbf{y}||^2$$

In this case, the only zero of p is given by

$$t_0 = \frac{-2 < \mathbf{x}, \mathbf{y} >}{2||\mathbf{y}||^2} = \frac{- < \mathbf{x}, \mathbf{y} >}{||\mathbf{y}||^2} \in \mathbb{R}.$$

Furthermore,

$$0 = p(t_0) = < \mathbf{x} + t_0\mathbf{y}, \mathbf{x} + t_0\mathbf{y} >$$

But by the positive definiteness property of the inner product we have that

$$0 = \mathbf{x} + t_0\mathbf{y} \Leftrightarrow \mathbf{x} = -t_0\mathbf{y}$$

Therefore, $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.